Transformer Riffage

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Abstract

Ever since the electrification of guitar, there has been an explosion in rhythmic complexity, tone experimentation and playing techniques. Music generation using Transformers Huang et al. (2018) has generally been used for creating mostly musical notes. In this paper, a new approach is proposed to capture the diverse expressions commonly featured in modern metal music. A new tokenisation method encapsulates the various guitar techniques and a sequence level attention component which tries to capture the segmentation between different musical phrases. An inference method is implemented that provides meaningful validation without human evaluation.

Keywords: Transformer models, Music, Generative, Sequence modelling, Attention mechanism

1 Introduction

A *riff* is a short, repeated motif or figure in the melody or accompaniment of a musical composition. Randel (1986) This type of composition is particularly prevalent in rock, metal, blues, funk and jazz. It is often the main feature that gives character and structure to a song. Recent developments in Transformers architecture have inspired many to generate sequences by treating musical notes as symbolic language.

Musical elements like notes, timestamps, and velocities are converted into discrete tokens similar to text processing. Suresh et al. (2020) A widely adopted method for representing musical notes digitally is the MIDI format, which encodes pitch, duration, and dynamics. Using such a format for musical expression has limitations like resolution and bandwidth constraints Shamoon (2024) when compared to richer representations like audio recordings and detailed scores. Modern guitar riffs frequently employ intense and abrupt expressive techniques, which contribute significantly to its overall identity. To accurately represent the nuanced expressive elements of modern music, a score-based format is preferable over simpler encodings like MIDI. A comprehensive list of these expressive elements is provided in Appendix A. Generating modern music demands capturing not only note patterns but also the associated expressions.

2 Encoding

The original Transformer model Vaswani et al. (2017) used a vocabulary of tokens derived from the input language corpus, with each token represented as an integer index. A note is represented by multiplying the guitar's fret number by the string number to generate a unique index. This approach not only helps the model learn the patterns of movement on the fretboard but also provides flexibility to accommodate alternative tunings, which are increasingly common in modern guitar playing. With 6 strings and 23 frets per string, the instrument offers 138 distinct fretboard positions. The rhythmic structure is divided into seven beat durations: whole, 1/2, 1/4, 1/8, 1/16, and 1/32 and 1/64 notes. These divisions provide varying levels of temporal granularity for representing note durations. Each beat can further be subdivided into 14 distinct rhythmic values which is listed in Appendix C, which makes a total to 82 unique beat indices. Since palm muting is a prominent feature in metal music, it is assigned a unique token in the representation. This encoding process produces 138 * 98 * 2 = 27048 unique musical tokens. The expressive vocabulary listed in Appendix A is represented using the same format. A unique integer is assigned to various accents used in guitar playing. The necessity of using relative positional encodings rather than absolute ones, is already shown to enable state reuse without causing temporal confusion Dai et al. (2019a). Since music is highly sequential such an encoding type should preserve timing, order, and spacing between notes.

Let L be the sequence length and d be the dimensionality of each attention head.

We define the relative positional embedding matrix:

$$\mathbf{R} \in \mathbb{R}^{(2L-1)\times d}$$

where the size $2L-1$ corresponds to all possible relative positions from $-(L-1)$ to $+(L-1)$.

The embedding lookup for a relative position $r = j - i$ (shifted by $L-1$ to be non-negative) is given by:

$$\mathbf{r}(r) = \mathbf{R}[r + (L-1)], \quad r \in \{-(L-1), \dots, 0, \dots, (L-1)\}$$

Here, $\mathbf{r}(r) \in \mathbb{R}^d$ is the embedding vector corresponding to relative position r.

2.1 Note representations

The formula for note encodings is given by:

$$note\_encodings = ni + bt * 276 + bb * 276 * 14 \tag{1}$$

$$ni = fn + sn * 23 + 6 * 23 * nval \tag{2}$$

$$nval = \begin{cases} 1 & \text{if, Palm Mute} \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

$$bt = \begin{cases} 1 + d & \text{if, Triplet(3)} \\ 2 + d & \text{if, Quintuplet(5)} \\ 3 + d & \text{if, Sextuplet(6)} \\ 4 + d & \text{if, Septuplet(7)} \\ 5 + d & \text{if, Nonuplet(9)} \\ 6 + d & \text{if, Undecuplet(11)} \end{cases} \tag{4}$$

$$d = \begin{cases} 7 & \text{if, Dotted note} \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

$$bb = \begin{cases} 0 & \text{if, } 1 \;/\; 64 \\ 1 & \text{if, } 1 \;/\; 32 \\ 2 & \text{if, } 1 \;/\; 16 \\ 3 & \text{if, } 1 \;/\; 8 \\ 4 & \text{if, } 1 \;/\; 4 \\ 5 & \text{if, } 1 \;/\; 2 \\ 6 & \text{if, } 1 \;/\; 1 \end{cases} \tag{6}$$

2.2 Accent representations

2.2.1 chord

A chord is a group of notes played together, typically three or more. Figure 1 depicts a C major chord comprising of 3 notes - root note [C] played on the fret 2 / 5th string , major third [G] on fret 1 / 4th string and perfect fifth [E] on fret 2 / 1st string.

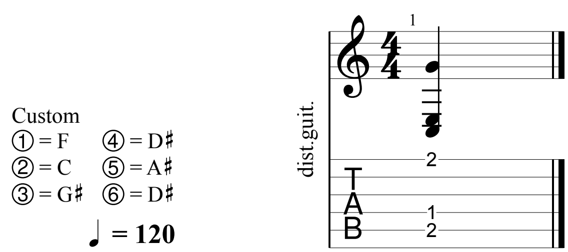

Figure 1: C major chord shape on a 6 string with custom tuning.

15413	27051	15389	27051	15321	27051
C	BARRE_NOTE	E	BARRE_NOTE	G	BARRE_NOTE

Table 1: C major chord quarter note tokenisation in a sequence.

2.2.2 other

Accentuations like BEND_NOTE, SLIDE_NOTE etc are represented by the note encodings from Section 2.1 followed by an accent token listed in Appendix A.

2.3 EOS and BAR

A music bar (also called a measure) is a fundamental unit in musical notation that divides music into segments of equal time, defined by the time signature. Each bar is separated by a BAR token. A collection of measures making a song is separated by an EOS token.

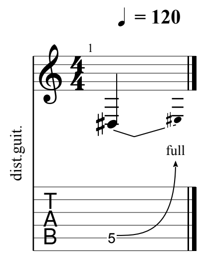

Figure 2: Fret 5 string 5 quarter note bend

15416	27052
C	BEND_NOTE_1

Table 2: Bend note tokenisation

3 Embedding

A bar consists of a fixed number of beats, composed of musical notes that were tokenised as described in the previous section. Each bar is treated like a sentence. The tokens representing the accents or expression in riff are placed next to each note in the sentence. An example of a riff and its tokenisation is given in Figure 2 . It has been demonstrated that the transformer equipped with relative attention is very well-suited for generative modelling of symbolic music. Huang et al. (2018). The chunking strategy used to carry over context between sequences during training follows the approach introduced in Yang et al. (2019), which builds upon Transformer-XL's segment-level recurrence mechanism. This method enables the model to capture longer-range dependencies by reusing hidden states from previous token chunks across segments.

4 Training

4.1 Model

The Transformer is built from a decoder-only stack with masked self attention as described in Huang et al. (2018). Although, segment-level recurrence mechanism is captured in Transformer-XL Dai et al. (2019b), its memory is limited to one previous segment at a time. To model dependencies across a whole batch, the gaussian distributions of the sequences are learned through a Multi layer Perceptron layer. If the input is a batch of sequences defined by:

$$X \in \mathbb{R}^{B \times T \times d} \tag{7}$$

Where:

- B: batch size (number of sequences)
- T: number of tokens per sequence
- d: embedding dimension
- $X_i \in \mathbb{R}^{T \times d}$: the i-th sequence in the batch

Sequence level mean pooling is performed by:

$$\bar{x}_i = \frac{1}{T} \sum_{t=1}^{T} X_{i,t} \quad \in \mathbb{R}^d \tag{8}$$

Gaussian parameters are then computed with a Multi Layer Perceptron. Let $f : \mathbb{R}^d \to \mathbb{R}^{2d}$ be a learned MLP. Then:

$$[\mu_i, \log \sigma_i] = f(\bar{x}_i) \quad \text{with} \quad \mu_i, \log \sigma_i \in \mathbb{R}^d \tag{9}$$

Ensure positivity of standard deviation:

$$\sigma_i = \exp(\log \sigma_i) + \varepsilon \quad \text{where } \varepsilon > 0 \tag{10}$$

Each sequence is then represented as a diagonal multivariate Gaussian:

$$\mathcal{N}_i = \mathcal{N}(\mu_i, \mathrm{diag}(\sigma_i^2)) \tag{11}$$

Vectorized for the batch:

$$[\mu, \log \sigma] = f(\bar{X}) \in \mathbb{R}^{B \times 2d} \tag{12}$$

$$\sigma = \exp(\log \sigma) + \varepsilon \in \mathbb{R}^{B \times d} \tag{13}$$

The pairwise KL divergence between each pair of Gaussians Zhang et al. (2023):

$$\mathrm{KL}(\mathcal{N}_i \parallel \mathcal{N}_j) = \frac{1}{2} \sum_{k=1}^{d} \left[\log \left(\frac{\sigma_{j,k}^2}{\sigma_{i,k}^2} \right) + \frac{\sigma_{i,k}^2}{\sigma_{j,k}^2} + \frac{(\mu_{i,k} - \mu_{j,k})^2}{\sigma_{j,k}^2} - 1 \right] \tag{14}$$

This produces a KL matrix:

$$\mathrm{KL\_matrix} \in \mathbb{R}^{B \times B}, \quad [\mathrm{KL\_matrix}]_{i,j} = \mathrm{KL}(\mathcal{N}_i \parallel \mathcal{N}_j) \tag{15}$$

A simpler version of the attention matrix from Dai et al. (2019a) is used from which the KL divergence-based sequence bias is subtracted:

$$\mathrm{RA}_{\mathrm{KL}} = \mathrm{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} + QR^\top - \alpha \cdot \mathrm{KL}_{\mathrm{bias}} \right) \tag{16}$$

A weighted sum over relative positional value contributions V_{rel} is then added:

$$\mathrm{Output} = \mathrm{RA}_{\mathrm{KL}} \times V + \mathrm{RA}_{\mathrm{KL}} \times V_{\mathrm{rel}} \tag{17}$$

4.2 Loss function

The model's loss is composed of three components:

$$Totalloss = CrossEntropy + Repetitionloss + Sequenceloss \tag{18}$$

4.2.1 Cross Entropy loss

Cross-entropy loss is the standard choice for training deep neural networks and logistic regression models for both binary and multi-class classification tasks. ml glossary (2017)

4.2.2 Repetition loss

Repetition helps organise musical ideas, making pieces more coherent and memorable. It gives listeners reference points and expectations, aiding in the recognition and recall of themes or motifs. Hu et al. (2022) Composers have long used stochastic (probability-based) methods to inject randomness into repeated motifs, sometimes even employing quantum randomness for truly unpredictable results. Yamada et al. (2021) This demands careful consideration, as it relies on the listener's inherent ability to recognise patterns and their natural inclination toward novelty. Since it is already known that tokens listed in Appendix A cannot repeat, it is penalised through repetition loss making them less likely to be selected again in subsequent steps. A variant of the repetition penalty introduced in Keskar et al. (2019) is employed; however, instead of greedy decoding, multinomial sampling is used to generate tokens to get semantically different output sequences Song et al. (2024). Including this loss during training eliminates the need to explicitly handle constraints for generating valid sequences at inference time

$$\text{Let } p_{b,t,v} = \text{softmax}\left(\frac{\text{logits}_{b,t,v}}{T}\right), \quad \text{for } b = 1, \dots, B;\ t = 0, \dots, T-1;\ v = 1, \dots, V \tag{19}$$

$$\tag{20}$$

Compute average past probabilities:

$$\bar{p}_{b,t,v} = \frac{1}{t} \sum_{k=0}^{t-1} p_{b,k,v}, \quad \text{for } t \geq 1 \tag{21}$$

Repetition score (dot product of current and average previous probabilities)

$$R_{b,t} = \sum_{v=1}^{V} p_{b,t,v} \cdot \bar{p}_{b,t,v}, \quad \text{for } t \geq 1 \tag{22}$$

Final penalty (mean repetition score across batch and sequence, excluding)

$$Repetitionloss = \frac{\lambda}{B(T-1)} \sum_{b=1}^{B} \sum_{t=1}^{T-1} R_{b,t} \tag{23}$$

where λ is a repetition penalty weight hyper parameter.

4.2.3 Sequence Penalty loss

Pairwise KL divergence between each pair of Gaussians is given by (14). Given a batch of B sequences, Sequence loss is defined by:

$$\mathcal{L}_{\text{KL}} = \frac{1}{B} \sum_{b=1}^{B} \text{KL}(\mathcal{N}_q^{(b)} \,\|\, \mathcal{N}_k^{(b)}) \tag{24}$$

To make the training more stable, the sequence loss is annealed and clamped.

$$Sequenceloss = \min\left(\frac{\text{step}}{\text{EPOCHS}} \cdot \left(\frac{\delta \cdot \mathcal{L}_{\text{KL}}}{\text{BATCH}}\right),\ 0.2\right) \tag{25}$$

where, δ is a sequence weight hyper parameter.

4.2.4 Dataset

Relavant statistics about the dataset used for training and evaluation are provided in Appendix (B)

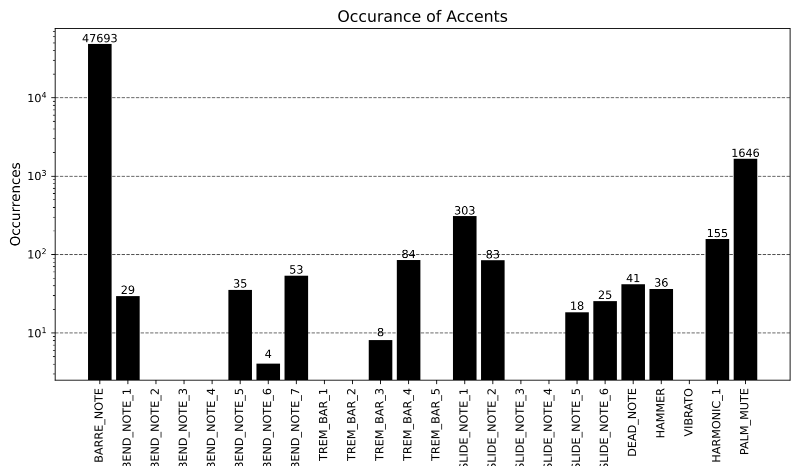

Figure 3: Accent occurrences in training data.

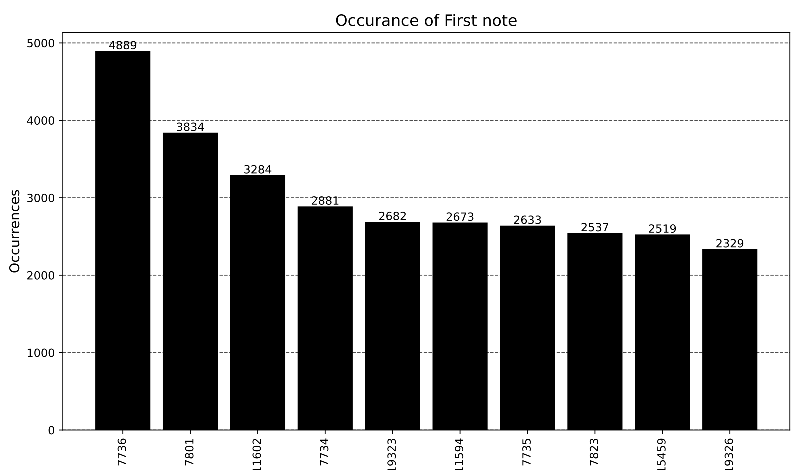

Figure 4: Token occurrences of first note of a musical measure in training data.

4.3 Grid search over Hyper-Parameters

To determine the optimal configuration, a grid search was performed over a predefined set of hyperparameter values. The search included combinations of key parameters such as learning rate, sequence length, model dimensionality, Each configuration was trained for 10 epochs, and performance was evaluated on a validation set using loss. The best-performing set of hyperparameters was selected based on the lowest validation score.

5 Inference

Given that the model is autoregressive and expects an initial token, selecting the start token based on the distribution of starting notes of musical measures in the training data yields more consistent generation.

The idea of context carryover has been well-established in Dauphin et al. (2016). During inference, the model carries over context from previous sequences by using the last token of the preceding measure as the first token for the subsequent measure. To compare the quality of predicted tokens (P) from true distributions (Q), relative entropy is calculated using Kullback-Leibler divergence:

$$D_{\mathrm{KL}}(P \parallel Q) = \sum_{x \in X} P(x) \log \left(\frac{P(x)}{Q(x)} \right) \tag{26}$$

The reported loss values were computed as the mean over five independent training runs after 100 epochs for hyper-parameter values listed in Appendix (D):

Loss	**Mean**	**Std. dev.**
Cross Entropy	1.6827	0.0437
Repetition	0.0323	0.0002
Sequence	0.1996	1e-6
Total	1.8021	0.1051
Validation	1.8021	0.1051

Table 3: Loss values

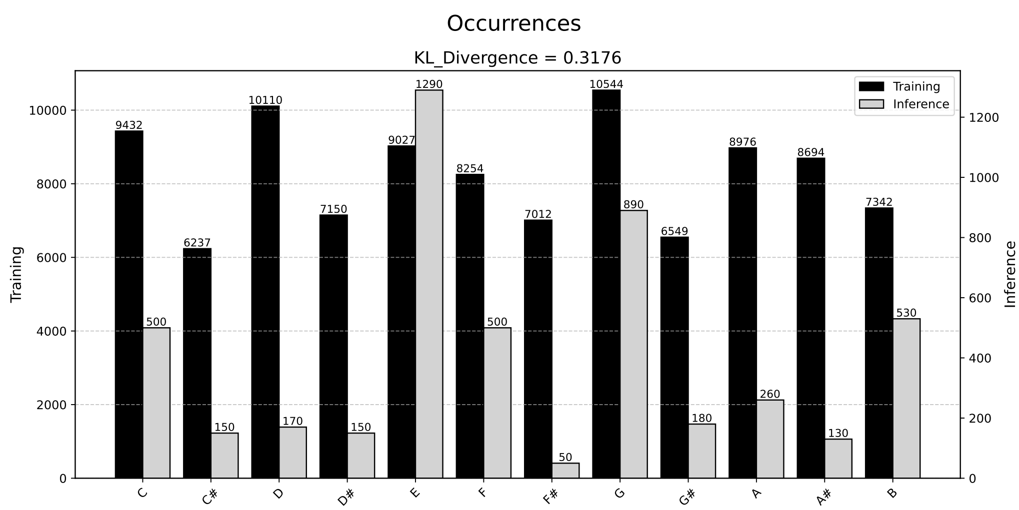

Figure 5: Note occurrences of training data and test data

The model produces musically pleasing sequences, effectively combining notes, beats, and accents while maintaining sequence context between measures. Evaluation was performed using a temperature setting of 0.7.

Figure 6: Test results showing 5 musical measures

Implementation details and hyper-parameter settings are provided in the associated GitHub repository Begari (2025a).

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Appendix A.

In this appendix a comprehensive list of accents played on the guitar is provided.

Accent	**Unique Index**
BAR	27049
BOS	27050
BARRE_NOTE	27051
BEND_NOTE_1	27052
BEND_NOTE_2	27053
BEND_NOTE_3	27054
BEND_NOTE_4	27055
BEND_NOTE_5	27056
BEND_NOTE_6	27057
BEND_NOTE_7	27058
TREM_BAR_1	27059
TREM_BAR_2	27060
TREM_BAR_3	27061
TREM_BAR_4	27062
TREM_BAR_5	27063
DEAD_NOTE	27064
SLIDE_NOTE_1	27065
SLIDE_NOTE_2	27066
SLIDE_NOTE_3	27068
SLIDE_NOTE_4	27069
SLIDE_NOTE_5	27070
SLIDE_NOTE_6	27071
HAMMER	27072
VIBRATO	27073
HARMONIC_1	27074
EOS	27075

Table 4: Example of unique indexing of guitar accents.

Appendix B.

This appendix explains relevant statistics about the dataset used for training and evaluation. The rhythm guitar tracks were obtained from Begari (2025b). The dataset consists files in .gp5 format in standard tuning. From the full set, a total of 6208 sequences were selected via random sampling for training. An 80/20 split was used to create the training and validation sets. For testing, top 10 first notes of each measure were selected as prompts for sequence generation and evaluation.

Appendix C.

In this appendix a comprehensive list of rhythmic patterns are listed.

Pattern	Unique Index
Base	1
Triplet	2
Quintuplet	3
sextuplet	4
septuplet	5
9 tuplets	6
11-tuplets	7
Dotted - Base	8
Dotted - Triplet	9
Dotted - Quintuplet	10
Dotted - sextuplet	11
Dotted - septuplet	12
Dotted - 9 tuplets	13
Dotted - 11-tuplets	14

Table 5: Example of unique indexing of rhythm.

Appendix D.

This appendix lists hyper-parameter configuration used to generate results.

Property	Value
FFN_HIDDEN	512
MAX_SEQ_LENGTH	16
NUM_HEADS	4
DROP_PROB	0.3
NUM_LAYERS	3
D_MODEL	512
LEARNING_RATE	0.0001
TEMPERATURE	1.0
OVERLAP	8
LAMBDA	0.5
ALPHA	0.1
DELTA	0.001

Table 6: Hyper-parameter values for training.

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